Quantum-Sound Factorization: On the Distribution and Structure of Irreducible Sets

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The Threat

- **According to the NSA, "the impact of adversarial use of a quantum computer could be devastating to [National Security Systems] and our nation."**
- The crux of most modern digital security is the difficulty of *integer factorization* – that is, factoring large integers into primes.
- However, quantum computers can solve this problem with alarming speed due to Shor's algorithm, hence threatening digital security.
- **It is hence vital that we find with an alternative.**

The Solution

- *Additive Combinatorics*, developed in the late 19th century, is now one of the most wide and active fields in mathematics.
- Just as primes are the building blocks of *natural numbers*, Additive Combinatorics studies factorization in the world of *sets*.
- In 2005, Kim and Roush proved that **the problem of set factorization over $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ is NP-hard classically and likely difficult for quantum computers as well**.
- **The goal is to build a cryptosystem based on set factorization, but we lack the mathematical understanding of the problem required.**
- **This work advances the state-of-the-art: resolving a key conjecture, uncovering a vulnerability, and advancing toward a full understanding of set factorization.**

Background

- Just as natural numbers have a complex factorization structure under multiplication, sets exhibit similar behavior under the *sumset* operation:
$$A + B = \{a + b : a \in A, b \in B\}.$$
- A set A is called *irreducible* if it does not factor further (i.e. there are no nontrivial sets B, C such that $A = B + C$), analogous to prime numbers.
- The problem of *set factorization* is determining how a set factors into irreducibles.
- Since set factorization in the natural numbers is quantum-secure, we focus on subsets of $\mathbb{N}_0$. We restrict ourselves to finite sets containing 0, as they retain all the complexity of set factorization.

Examples

- We have that $\{0, 1\} + \{0, 1, 2\} = \{0, 1, 2, 3\}$.
- The set $\{0, 1, 3\}$ is irreducible.
- One factorization of $\{0, 1, \dots, 9\}$ is $3\{0, 1\} + 2\{0, 1, 3\}$. Note that this is not unique!

- These sets under the sumset operation form a *monoid* (a group without inverses), denoted by $\mathcal{P}_{fin,0}(\mathbb{N}_0)$.
- The study of irreducible sets is of interest in many other domains, such as idempotent analysis and random number generation.
- Below is a visualization of the sumset operation.

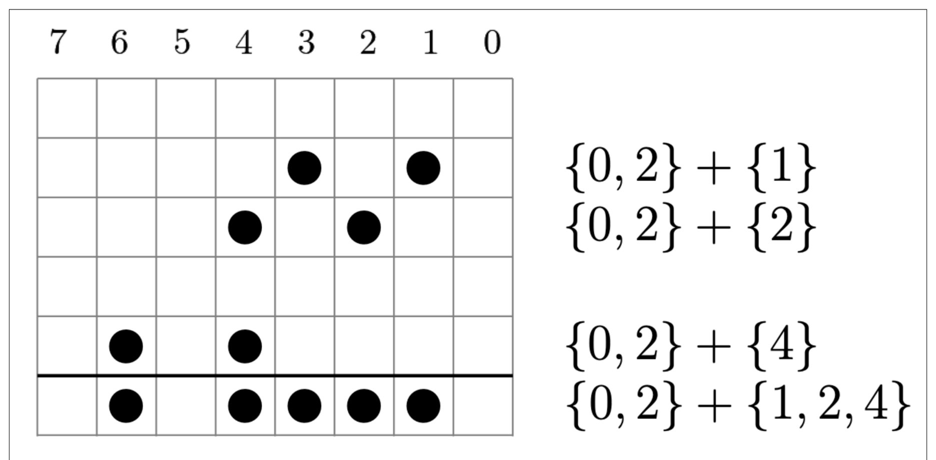


Figure 1: Graphical Representation of $\{0, 2\}+\{1, 2, 4\} = \{1, 2, 3, 4, 6\}$, due to Gal Gross (modified)

Solving a Key Open Conjecture

The study of prime numbers has fascinated mathematicians since Ancient Greece, driving some of the most fundamental questions in mathematics: *How many primes exist within a given range? What structural properties define them?* These inquiries have shaped entire fields of research, from number theory to cryptography. A striking example is the **Riemann Hypothesis (RH)** – a famous unsolved problem in mathematics, with a **$1,000,000 prize** for its resolution.

To develop a secure cryptosystem based on **irreducible sets**, we must answer analogous questions about their structure and distribution. The natural starting point is determining *how many irreducible sets exist.* For prime numbers, this question is elegantly addressed by the **Prime Number Theorem**, one of the most celebrated results of the 19th century. In our setting, we define α_n as the number of irreducible subsets of $\{0, 1, \dots, n\}$, and $\alpha_{n,k}$ as the number of such subsets of size k. The strongest existing bounds, due to **Geroldinger and Bienvenu**, establish that $\alpha_n = 2^n(1 - \exp(\Omega(n)))$, meaning that the probability of a random set *not* being irreducible is exponentially small. However, no comparable understanding exists for $\alpha_{n,k}$ – a major gap in our knowledge.

We resolve a key open conjecture providing a complete characterization of how the size of a set influences its irreducibility. These results mark a **breakthrough** in our understanding of set factorization, laying the foundation for practical cryptographic applications.

Theorems (A., 2025), (A.-Gotti-Lu, 2024)

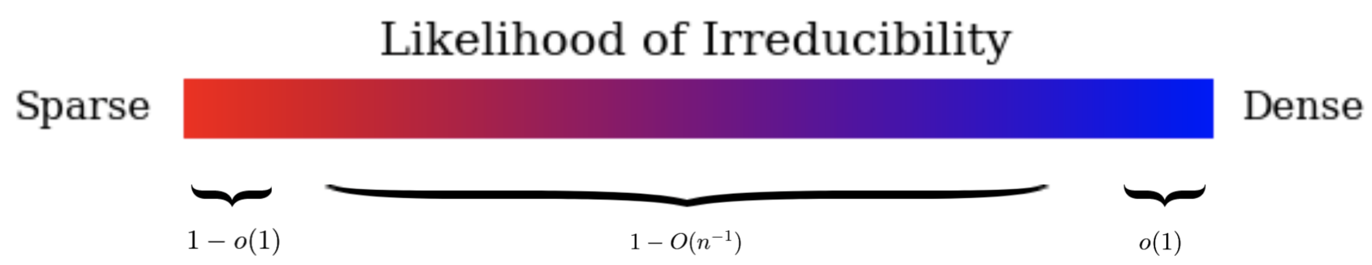
Let $k(n) : \mathbb{N} \to \mathbb{N}$. We have the following:

- If $k(n) = o(n)$, $\alpha_{n,k(n)} = \binom{n}{k(n)-1}(1 - o(1))$.
- If $k(n)/n \to \delta$ as $n \to \infty$, where $0 < \delta < 1$, then $\alpha_{n,k(n)} = \binom{n}{k(n)-1}(1 - O(n^{-1}))$.
- If $k(n) = n - o(\sqrt{n})$, then $\alpha_{n,k(n)} = o(1)\binom{n}{k(n)-1}$.

In other words, we prove the following (with precise asymptotics).

- Sparse sets are highly likely to be irreducible.
- Sets that are dense but not too dense are very highly likely to be irreducible.
- Sets that are exceedingly dense are highly unlikely to be irreducible.

For example, a set like $\{0, 1, 19, 43\}$ is a lot more likely to be irreducible than a set like $\{0, 1, 2, 3, \dots, 43\} \setminus \{0, 17, 32\}$. The below graphic represents this collection of theorems visually. Note that $o(1)$ means a function tending to 0, and $O(n^{-1})$ just means a function tending to 0 at a faster rate than $f(x) = x^{-1}$.



This is crucial information for a cryptosystem, because it tells us exactly where to find irreducibles.

A Vital Irreducibility Criterion

Ensuring the security of a cryptosystem is of utmost importance. Here, security hinges on the difficulty of understanding set factorization. While set factorization is likely difficult for quantum computers to crack deterministically, we must also **mitigate probabilistic attacks**, which exploit structural patterns to break encryption.

The best-known structural property previously applied to *almost no* irreducible sets (0%). **We discover an alarming property shared by $6 - 18\%$ of all sets, posing a critical vulnerability. Cryptosystems in development must be modified to mitigate this risk.** **No such structural weakness has ever been found in any other domain—not even among the rational primes.**

The Criterion

A set S is irreducible if:

- $2m \notin S$, where $m = \min S_{>0}$.
- For all $s \in S_{>m}$, there is a $t \in S_{<s}$ such that $s + t \notin S$.

We call such a set *strongly irreducible*. Note that this property is polynomial-time verifiable.

Examples

- The set $S = \{0, 1, 3\}$ is strongly irreducible because $2 \cdot 1 \notin S$ and $3 + 1 \notin S$.
- The set $S = \{0, 1, 3, 5, \dots, 2n + 1\}$ is strongly irreducible for all n because $2 \cdot 1 \notin S$ and for all $s \in S_{>1}$, we can choose $t = 1$ and have $s + t \notin S$.

The following theorem establishes the aforementioned $6 - 18\%$ result.

Theorem (A., 2025)

Let s_n be the number of strongly irreducible subsets of $\{0, 1, \dots, n\}$. Then
$$6.38\% \approx \frac{49}{768} \le \lim_{n\to\infty} \frac{s_n}{2^n} \le \frac{3}{16} = 18.75\%.$$

We additionally prove that almost all sparse sets (sets with $k \ll n/\log n$, in particular) are strongly irreducible.

Additional Results

- We establish additional results about irreducible sets, of interest to pure mathematicians.
- An interesting result is that irreducible sets approximately follow a binomial distribution. For each N, one can define a natural random variable X_N with probability mass function
$$\mathbb{P}(X_N = k) = \frac{\alpha_{N,k}}{\alpha_N}.$$

Then we have the following:

Theorem (A.-Gotti-Lu, 2024)

For each $N \in \mathbb{N}$, let $Y_N \sim \mathrm{Bin}(N, 1/2)$. Then for every $r \in \mathbb{N}$ we have that the rth-moments of X_N and Y_N are roughly the same:
$$\lim_{N\to\infty} \frac{\mathbb{E}(X_N^r)}{\mathbb{E}(Y_N^r)} = 1.$$
In other words, the moment generating functions of X_N and Y_N are asymptotically equivalent.

- This behavior can be observed already for $N = 9$:

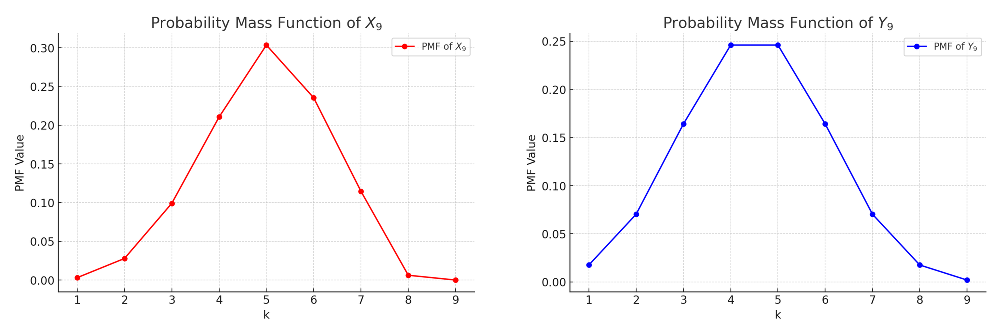


Figure 2: Side-by-side distributions of X_9 and Y_9

- We prove additional algebraic properties of $\mathcal{P}_{fin,0}(\mathbb{N}_0)$ and other related monoids. In particular, we **generalize results of Geroldinger & Bienvenu** on primes in such monoids.

Implications & Future Work

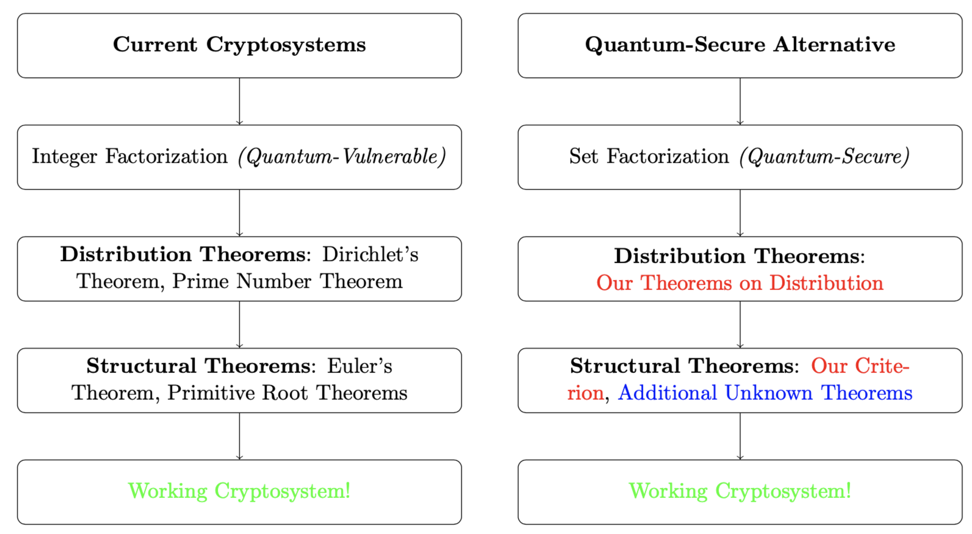


Figure 3: Over-simplified flowchart detailing progress towards a quantum-secure cryptosystem

- **Without inventing quantum-secure cryptography, all modern encryption could collapse. This research brings us closer to a quantum-secure cryptosystem.**
- We surpass a few major barriers towards creating an efficient, quantum-secure cryptosystem:
- **We resolve a major open conjecture, providing key insight as to where we may find irreducibles.**
- **By proving this conjecture, we not only deepen our understanding of set factorization but also take a decisive step toward constructing a quantum-secure cryptosystem.**
- **We discover a key vulnerability that must be mitigated in such cryptosystems.**
- Additionally, we prove results of interest to pure mathematicians about irreducible sets.
- In the future, the goal is to further improve asymptotics for $\alpha_{n,k}$ and s_n.
- We also wish to further understand the divisor function on sets, which seems to be the final barrier towards said cryptosystem. This divisor function also tells us about the entropy of random number generators based on sumsets.